Class 2: Probability and Events

STA 220
Mai Ghannam
Summer 2026

Overview

This class will cover Module 2

- Module 2: Probability Events

Topics include:

- Probability and Events
- Conditional Probability
- Independence of Events
- Bayes' Rule

Randomness and Probabilities

Randomness and Probability

Randomness and Probability

- Randomness is why probability is central in statistics.
- We cannot know with certainty what will happen in a single experiment.
- If we could, there would be no need to run the experiment.
- However, we can often describe which outcomes are more likely than others.

Randomness and Probability

Example (fair coin):

- We cannot predict whether a single flip is heads or tails.
- But we know both outcomes are equally likely.

Example (biased coin):

- We still cannot predict a single outcome.
- But we know heads is more likely than tails.

Key Idea

•Individual outcomes are uncertain.

•But over many repetitions, outcomes follow a stable and predictable pattern.

Examples of Randomness

- The result of rolling a 6-sided die
- The time that a bus arrives at the bus stop
- The height of the next person who walks into the room
- The sex of the next baby born in a hospital
- The maternal mortality rate in Canada next year

Probability and Outcomes

Probability is how we describe the chance that an outcome occurs in an experiment.

Example: A fair coin has a 50% chance of landing on heads.

Important:
This does *not* mean outcomes must balance in a small number of trials.

•Example: Two coin flips do not guarantee one head and one tail.

Probability Distributions

A probability distribution describes the chances of *all possible outcomes* of an experiment.

It tells us what we should expect **in the long run**, over many repetitions.

Example: For a fair coin, the distribution assigns:

- 50% to heads
- 50% to tails

- Think of it as a model of what happens if we repeat the experiment “forever.”

Events in Probability

When we run an experiment, we may be interested in specific outcomes.

Example: flipping two coins and getting **first a head, then a tail**

More often, we are interested in a **collection of outcomes** rather than a single one.

Example: flipping two coins and getting **at least one head**

Events

An **event** is a subset of all possible outcomes of an experiment.

We usually denote events using **capital letters**.

- Example:
  - $A =$"getting at least one head in two coin flips"

Probability of an Event

We write $P(A)$to denote the probability that event A occurs.

For probability to be meaningful, it must satisfy:

$$0 \leq P(A) \leq 1$$

Interpretation:

$P(A) = 1$:event A is certain to occur

$P(A) = 0$:event A will never occur

Sample Space and Events

A **sample space** is the set of all possible outcomes of a random experiment. It is usually denoted by S.

Events are subsets of the sample space.

Sample Space and Events

Example: Rolling a 6-sided die

- Sample space:
- $S = \{1,2,3,4,5,6\}$

Probability distribution (fair die):

- $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$

Examples of Events

- **Simple event:**

$$A = \{1\}$$

- **Compound event:**

$$A = \{1, 2, 3\}$$

Key Properties

- $0 \leq P(A) \leq 1$for any event A
- $P(S) = 1$

Sample Space: Die + Coin

Suppose we roll a die and then flip a coin.

The sample space consists of all possible combinations of outcomes.

There are 12 possible outcomes:

$$S = \{(1\mathrm{H}), (1\mathrm{T}), (2H), (2\mathrm{T}), (3H), (3\mathrm{T}), (4H), (4\mathrm{T}), (5H), (5\mathrm{T}), (6H), (6\mathrm{T})\}$$

How do we count outcomes?

When experiments are combined, we can often use multiplication:

Number of outcomes:

$$6\ \textit{(die outcomes)} \times 2\ \textit{(coin outcomes)} = 12$$

Key Idea

The size of the sample space is the number of all possible combinations of outcomes.

Product Rule (Counting Principle)

If one experiment can occur in m ways and a second independent experiment can occur in n ways, then the total number of combined outcomes is:

$$m \times n$$

Product Rule (Counting Principle)

If one experiment can occur in m ways and a second independent experiment can occur in n ways, then the total number of combined outcomes is:

$$m \times n$$

Product Rule (Counting Principle)

Suppose my experiments has three parts now:

- Roll a six-sided die
- Flip a coin
- Spin a colour in five-color spinner (equal parts)

What is the total number of events in S?

Complement of an Event

Sometimes we are interested in the probability that an event **does not occur**.

The **complement** of an event A, written A^c ,represents all outcomes where A does not happen.

Key Formula

$$P(A^c) = 1 - P(A)$$

Complement of an Event

Example

Let A= “it rains tomorrow”

Then:

$P(A)$ =probability it rains

$P(A^c)$ =probability it does **not** rain

Suppose:

$$P(A) = 0.3$$

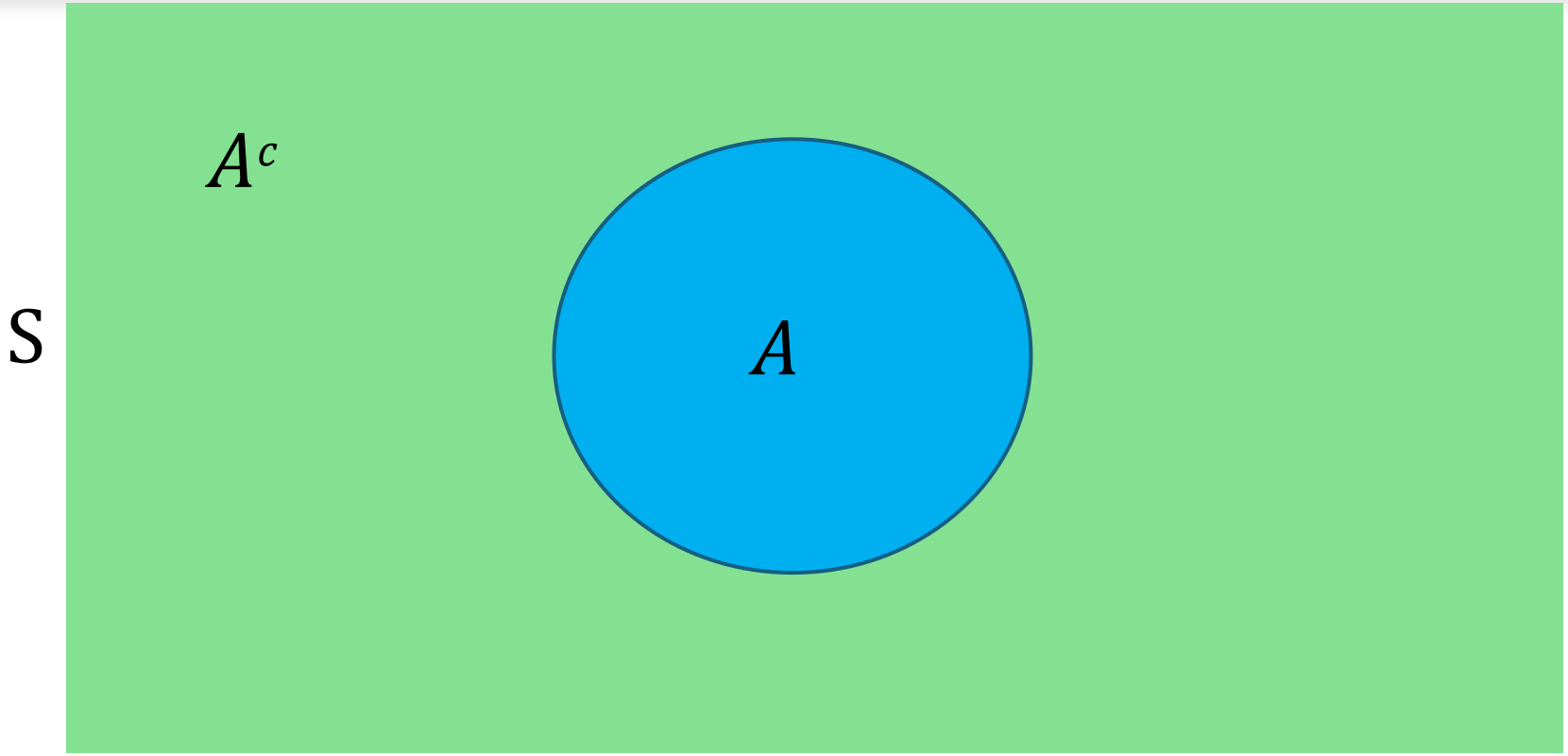
Then:

$$P(A^c) = 1 - 0.3 = 0.7$$

Key Idea

The probability of an event and its complement always add to 1.

Complement of an Event Venn Diagram



Addition Rule

Sometimes we want the probability of **combinations of events**.

Key Rule

$$P(A\ or\ B) = P(A) + P(B) - P(A\ and\ B)$$

What do "or" and "and" mean?

A or B means:

- A happens, or
- B happens, or
- both A and B happen

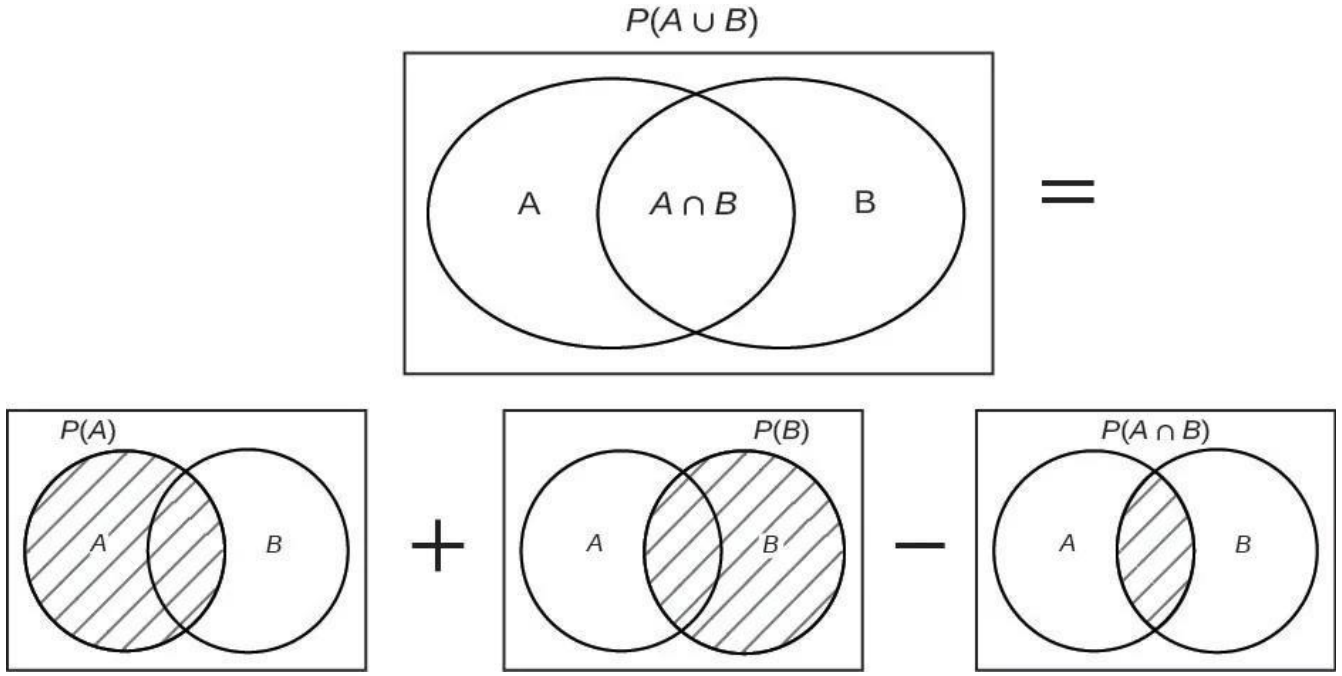
A and B means:

- both events happen at the same time

Addition Rule

Why do we subtract $P(A\ and\ B)$?

- Because the overlap (both A and B) is counted twice when we add $P(A)$and $P(B)$.



Check Your Understanding!

What happens when I try to use the Addition Rule and events A and B have no outcomes in common?

A. I can't compute P(A or B)
B. I can't compute P(A and B)
C. I get that P(A or B) = 0
D. I get that P(A and B) = 0

Example: Rolling 2 Dice

When rolling two dice, each die has 6 outcomes.

- So the total number of possible outcomes is:

$$6 \times 6 = 36$$

Possible Sums

- Instead of looking at ordered pairs, we often look at the **sum** of the two dice.

The possible sums are:

$$2,3,4,5,6,7,8,9,10,11,12$$

Example: Rolling 2 Dice

Key Idea

- Different outcomes can produce the same sum.
- Example:

  $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ all give sum = 7

Sum of 2 Dice Chart

	1	**2**	**3**	**4**	**5**	**6**
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Sum	2	3	4	5	6	7	8	9	10	11	12
# of ways to get it											
Prob											

Example: Rolling 2 Dice

Example: Sum of Two Dice is NOT 6

When rolling two dice, there are:

$$36 \ \textit{equally likely outcomes}$$

The possible sums are:

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Event of Interest

Let A= "the sum is not 6"

Then:

$$A = \{2,3,4,5,7,8,9,10,11,12\}$$

Example: Rolling 2 Dice

Using the Complement Rule

It is easier to use the complement:

Event A^c" =the sum is 6"

There are 5 ways to get a sum of 6:

$$(1,5), (2,4), (3,3), (4,2), (5,1)$$

So:

$$P(A^c) = \frac{5}{36}$$

Final Answer

$$P(A) = 1 - P(A^c) = 1 - \frac{5}{36} = \frac{31}{36}$$

Key Idea

Sometimes it is easier to compute the complement first and subtract from 1.

Example: Rolling 2 Dice

Find the following probability that the sum of the 2 dice is at least 4.

Example: Rolling 2 Dice

Example: Combining Events (Two Dice)

Suppose we want:
(A) sum is **not 6**
(B) sum is **at least 4**

We want:

$$P(A \text{ or } B)$$

Conditional Probability

Conditional Probability

Conditional probability is closely related to contingency tables and relationships between two categorical variables.

Last week, when we looked at **conditional distributions**, we focused on one group at a time by restricting our attention.

This means we “ignored” part of the table to focus on a specific subgroup.

	Coffee-No	**Coffee-Yes**	**Total-Tea**
Tea-No	44	29	73
Tea-Yes	40	45	85
Total-Coffee	84	74	158

Conditional Probability

Example (from contingency tables)

- Among **non-coffee drinkers**, what proportion drink tea?
- Here we are only looking at:
  - the subgroup of non-coffee drinkers
  - and then studying outcomes *within that group*

Key Idea

- Conditional probability means we are finding probabilities **given that something is already known**.

	Coffee-No	**Coffee-Yes**	**Total-Tea**
Tea-No	44	29	73
Tea-Yes	40	45	85
Total-Coffee	84	74	158

Conditional Probability

Example (from contingency tables)

- Among **non-coffee drinkers**, what proportion drink tea?
- Here we are only looking at:
  - the subgroup of non-coffee drinkers
  - and then studying outcomes *within that group*

Key Idea

- Conditional probability means we are finding probabilities **given that something is already known**.

	Coffee-No	**Coffee-Yes**	**Total-Tea**
Tea-No	44	29	73
Tea-Yes	40	45	85
Total-Coffee	84	74	158

Conditional Probability

When we compute a conditional probability, we are working with two events:
A: the event we are interested in

- e.g. “tea drinkers”

B: the event that has already occurred (we condition on it)

- e.g. “non-coffee drinkers”

Key Idea
Since we know B has happened, we ignore everything outside of B.
We only consider outcomes that lie within B.

Conditional Probability

Venn Diagram Intuition
If we are inside the circle B, we ask:
What is the probability of also being in A?
Think of B as the **new sample space**.

Conditioning on B
We restrict our attention to outcomes in B.
This means removing all outcomes not contained in B.

Key Takeaway
Conditional probability = probability of A **within the reduced sample space** B

Conditional Probability Notation

Definition of Conditional Probability
The conditional probability of A given B is written as:

$$P(A \mid B)$$

Key Idea
When we condition on B, we restrict attention to outcomes where B has occurred.
So we are asking:

"Out of all outcomes in B, what fraction are also in A?"

Conditional Probability Notation

Two-Step Intuition
First, consider outcomes in both A and B: $A \cap B$
Then compare them to all outcomes in B

General Form

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Key Takeaway
Conditional probability is:
probability of the overlap $A \cap B$, **relative to** B

We are no longer using the full sample space — only B

Conditional Probability Example: Coin Flipping

Suppose we flip a coin 3 times.
The sample space has 8 equally likely outcomes:

$$S = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{THT}), (\mathrm{HTT}), (\mathrm{TTH}), (\mathrm{TTT})\}$$

Event of Interest
Let A= "two or more heads"
Then:

$$A = \{(\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{HHH})\}$$

So there are 4 outcomes in A.

Conditional Probability Example: Coin Flipping

Probability

$$P(A) = \frac{4}{8} = \frac{1}{2}$$

Key Idea

Probability = (number of favourable outcomes) ÷ (total number of outcomes)

All outcomes are equally likely in this experiment.

Conditional Probability Example: Coin Flipping

Flip a coin 3 times. The sample space is:

S= $\{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{THT}), (\mathrm{HTT}), (\mathrm{TTH}), (\mathrm{TTT})\}$

Let:

- A= “two or more heads”
- B= “heads on the first toss”

Conditional Probability Example: Coin Flipping

Conditioning on B

If we know the first toss was heads, we remove all outcomes that do not begin with H.

Our new sample space becomes:

$$B = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{HTT})\}$$

Which outcomes are also in A?

Outcomes with at least two heads:

$$A \cap B = \{(\mathrm{HHH})\}, (\mathrm{HHT}), (\mathrm{HTH})\}$$

There are:

3 favourable outcomes

4 total outcomes in the new sample space

Conditional Probability Example: Coin Flipping

Conditional Probability

$$P(A \mid B) = \frac{3}{4}$$

Key Idea

Once we condition on B, we no longer use all 8 outcomes.
We only work within the reduced sample space defined by B.

Conditional Probability
Example: Back to Coin Flipping

Using the Conditional Probability Formula

The sample space is:

S= $\{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{THH}), (\mathrm{THT}), (\mathrm{HTT}), (\mathrm{TTH}), (\mathrm{TTT})\}$

Let:

A= "two or more heads"

B= "heads on the first toss"

Step 1: Find the required probabilities

Outcomes in both $A$and B:

$$A \cap B = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH})\}$$

So:

$$P(A \cap B) = \frac{3}{8}$$

Outcomes in B:

$$B = \{(\mathrm{HHH}), (\mathrm{HHT}), (\mathrm{HTH}), (\mathrm{HTT})\}$$

So:

$$P(B) = \frac{4}{8}$$

Conditional Probability
Example: Back to Coin Flipping

Step 2: Apply the formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{3/8}{4/8} = \frac{3}{4}$$

Key Idea

This gives the same result as restricting the sample space to B then finding how many remaining outcomes are in A

Conditional Probability
Example: Back to Coin Flipping

Step 2: Apply the formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{3/8}{4/8} = \frac{3}{4}$$

Key Idea

This gives the same result as restricting the sample space to B then finding how many remaining outcomes are in A

Conditional Probability
Example: Back to Coin Flipping

Conditional Probability Using a Contingency Table

We can also analyze this problem using a contingency table.

Let:

the **columns** represent whether outcomes are in A or not in A

the **rows** represent whether outcomes are in B or not in B

	A=2 or more heads	Not A=1 or fewer heads	Row Total
B= head on first toss			
Not B=tail on first toss			
Column Total			8/8=1

Check Your Understanding!

TRUE or FALSE

If events A and B are disjoint (have no outcomes in common), then

$$P(A \mid B) = P(A)$$

Independent vs. Mutually Exclusive Events

Independence of Events

Independence of Events
Two events are called **independent** if knowing that one event occurred does not change the probability that the other event occurs.

Example: Roll a Die and Flip a Coin
Let:

- A= “roll a 3”
- B= “land on heads”

Question
Does knowing the coin landed on heads give us any additional information about whether the die roll is a 3?
The answer is **no**.
The coin flip does not affect the die roll, so the events are independent.

Key Idea
Independence is about information.
If knowing one event occurred does **not** change the probability of the other event, then the events are independent.

Independence of Events Definition

We can check whether two events A and B are independent in two equivalent ways:

1. Using Conditional Probability

Events A and B are independent if:

$$P(A \mid B) = P(A)$$

This means:

Knowing that B happened does **not change** the probability of A

Similarly:

$$P(B \mid A) = P(B)$$

Knowing A happened does **not change** the probability of B

2. Using the Multiplication Rule

Events A and B are independent if:

$$P(A \cap B) = P(A)\,P(B)$$

Independence of Events: Connection Between the Two Definitions

Independence of Events: Example (Rolling Dice/Coin Toss)

Consider:

$A = \{roll\ a\ 3\}$

$B = \{flip\ a\ head\}$

Sample Space:

S={(H,1), (H,2), (H,3), (H,4), (H,5), (H,6), (T,1), (T,2), (T,3), (T,4), (T,5), (T,6)}

Is P(A) = P(A|B) ?

	A=roll a 3	Not A= did not roll a 3	Probability
B=flip a head			
Not B=flip a tail			
Probability			

Mutually Exclusive Events

Another important idea is **mutually exclusive events**.

Two events are mutually exclusive if:
They have **no outcomes in common**

This is also called **disjoint events**.

Key Idea
The difference between **independence** and **mutual exclusivity** lies in the overlap:

$$P(A \cap B)$$

Mutually Exclusive vs Independent Events

1. Independent Events

- Events **can occur together**
- There may be overlap between A and B
- The overlap satisfies:

$$P(A \cap B) = P(A)P(B)$$

Common outcomes are possible

2. Mutually Exclusive Events

Events have **no overlap**

They cannot happen at the same time

$$P(A \cap B) = 0$$

No common outcomes at all

Mutually Exclusive vs Independent Events

Type	**Overlap ($(\mathbf{A} \cap \mathbf{B})$)**	**Interpretation**
Independent	Possible	Occur together, but unrelated
Mutually exclusive	Empty set	Cannot occur together

Mutually Exclusive Events Venn Diagram

Check Your Understanding!

Question: Can mutually exclusive events ever be independent?

Example: Blood Types

According to Canadian Blood Services:

Type O: 46% **Type A:** 42% **Type B:** 9% **Type AB:** remaining 3%

For one randomly selected person:

Let

$$A = \{\textit{person has Type A blood}\}$$
$$B = \{\textit{person has Type B blood}\}$$

Are A and B mutually exclusive, independent, or neither?

Example: Blood Types

We now consider **two different individuals**.

Let:

$$A^* = \{\textit{first person has Type A blood}\}$$
$$B^* = \{\textit{second person has Type B blood}\}$$

Question

Are $A^*$and B^* mutually exclusive , independent or neither?

Example: Blood Types

According to Canadian Blood Services:

Type O: 46% **Type A:** 42% **Type B:** 9% **Type AB:** remaining 3%

<u>Among 4 potential donors:</u>

a) What is the probability that they are all Type O?
b) What is the probability no one is Type AB?

Check your understanding!

Info 1: The probability of event A occurring is 0.3
Info 2: The probability of event C not occurring is 0.78
Info 3: The probability of event A and C occurring is 0.2

a) What is the probability that event A occurs if we know C occurred?

b) What is the probability that event C occurs if we know A occurred?

c) Are events A and C independent?

d) Ignore Info 2. If they were independent, what would be the value of P(C)?

Bayes' Rule

Bayes' Rule

Bayes' Rule is a useful result that lets us:

- use conditional probabilities we already know
- and **reverse them** to find a probability of interest

It is a **direct extension of the definition of conditional probability**.

It allows us to "flip" conditioning from

$$P(A \mid B)$$

to

$$P(B \mid A)$$

In many real problems:

$P(A \mid B)$

is easy to think about or given but we actually need

$P(B \mid A)$

Bayes' Rule connects the two.

Tree diagrams

Tree diagrams are a useful tool for working with **conditional probabilities**.

They help us:
- organize outcomes step-by-step
- keep track of all possible combinations of events

How to Use a Tree Diagram
Multiply along branches (left to right)
Move across the tree
Multiply probabilities along each path

$$P(path) = multiply\ along\ branches$$

Add outcomes in the same column (vertical grouping)
Combine probabilities of different paths leading to the same event
Add probabilities within the same category

$$P(event) = add\ relevant\ paths$$

Bayes' Rule and Tree Diagrams Example

We look at the probability structure for two children.

First Birth

Probability first child is **Female (F)**:

$$P(F_1) = 0.5$$

Probability first child is **Male (M)**:

$$P(M_1) = 0.5$$

Second Birth (Conditional)

If the first child is **Female**:

$$P(F_2 \mid F_1) = \frac{1}{3}$$

If the first child is **Male**:

$$P(F_2 \mid M_1) = \frac{2}{5}$$

Suppose we want to know the probability of a female first child given that we had a female second child: $P(F1|F2)$.

Interpreting Baye's Rule

From the tree diagram, we can rewrite the components of Bayes' Rule.

Numerator

$$P(A \cap B)$$

This is the probability that **both** A **and** B **occur**.

Denominator

$$P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$$

This can be rewritten using the law of total probability:

$$= P(A \cap B) + P(A^c \cap B)$$
$$= P(B)$$

So Bayes' Rule becomes:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Final Conclusion

Bayes' Rule is simply:
another way of writing the definition of conditional probability

It follows directly from:
- the definition of conditional probability
- and the law of total probability

Example: Genetic Testing for Thrombosis

A genetic test is used to determine whether a person has a predisposition for thrombosis, a condition involving blood clots that can block blood flow. It is known that approximately 3% of the population has this predisposition.

The test is highly accurate: if a person has the predisposition, the test correctly returns a positive result 99% of the time. If a person does not have the predisposition, the test correctly returns a negative result 98% of the time.

The question of interest is: if a person tests positive, <u>what is the probability that they actually have the predisposition?</u>

Lecture Summary

- In this lecture, we covered both parts of Module 2: Probability Events.

- **Practice Problems and Solutions are available on Quercus**
  - See Class 2 Course Materials on Quercus)

Looking Ahead

- **Quiz 1 (and bonus AI usage survey) is due on May 14 at 11:59 PM** (opens on May 11 at 12:00 AM)

- Class 3 (Tuesday) we will cover Random Variables